Consumer Technology Association™

CTA Standard

Improvements to CTA-861-I

CTA-861.7

June 2024

Consumer Technology Association™

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(Formulated under the cognizance of the CTA **R4 Audio and Video Systems Committee**.)

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FOREWORD

This standard was developed by the Consumer Technology Association's R4 Video Systems Committee.

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TABLES

Table A - Numbering changes of Tables 2

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Improvements to CTA-861-I

1 SCOPE

CTA-861.7 specifies changes to CTA-861-I [A].

The requirements of this standard are in addition to and complement CTA-861-I. All devices compliant to CTA-861.7 shall also comply with CTA-861-I, except that this standard modifies Sections 7.4, 7.5.14.1 through 7.5.14.4, and 7.5.19 of CTA-861-I.

2 REFERENCES

2.1 Normative References

The following standards contain provisions that, through reference in this text, constitute normative provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards listed here.

2.2 Normative Reference List

A. CTA-861-I, *A DTV Profile for Uncompressed High Speed Digital Interfaces*, February 2023, https://cta.tech/standards.

3 COMPLIANCE NOTATION

CTA defines the following compliance terms for use in its documents:

shall	This word indicates specific provisions that are to be followed strictly (no deviation is permitted).
shall not	This phrase indicates specific provisions that are absolutely prohibited.
should	This word indicates that a certain course of action is preferred but not necessarily required.
should not	This phrase means a certain possibility or course of action is undesirable but not prohibited.
may	This phrase indicates that a certain course of action is optional.
need not	This phrase indicates that a certain course of action is not required (i.e., optional), and this document does not express a recommendation as to preference.

4 TABLES AND FIGURES

This document adds and removes Tables in CTA-861-I [A]. This requires re-numbering Tables throughout the document. Table A provides an overview of Table numbers in CTA-861-I before and after the modifications by this document. Existing Tables of CTA-861-I are re-numbered according to Table A before any other modifications by this document:

Table A - Numbering changes of Tables

Section	Table Number in CTA-861-I	Table Number after applying this document
4 – 7.5.5	1 – 83	1 – 83
7.5.5	84	N/A
7.5.6 – 7.5.19.3	85 – 122	84 – 121
7.5.20	N/A	122
7.6 – Annex V	123 – 206	123 – 206
Annex V	N/A	207
Annex V	207	208

5 DOCUMENT TITLE

The title of the CTA-861 specification is modified, as shown below:

A DTV Profile for High-Speed Digital Interfaces

6 EDID AND VIDEO TIMINGS

The AVI definition in Section 2.2 of CTA-861-I [A] is modified, as shown below:

> **Auxiliary Video Information (AVI) InfoFrame:** Additional information (defined in CTA-861) related to the video being sent from a Source to a Sink.

The fourth paragraph of Section 4 of CTA-861-I [A] is modified, as shown below:

> If a Dual-Aspect Ratio DTV is receiving a Video Format Timing for which it has declared support for both Picture Aspect Ratios in EDID and the Source has indicated the Picture Aspect Ratio by including the AVI InfoFrame in the video stream, then the DTV shall display the Picture in the aspect ratio that has been indicated by the Source in the AVI InfoFrame. If the Source does not support transmission of the AVI InfoFrame and the Source supports both of the Dual-Aspect Ratio Video Timing Formats for a particular Video Timing defined by the Sink, then the Source shall provide the video to the DTV in the Preferred Picture Aspect Ratio.

The fifth paragraph of Section 6 of CTA-861-I [A] is split into two paragraphs, as shown below:

> DVI does not support the transmission of any InfoFrames, independent of CTA Extension version number.
>
> Sinks with a VSDB indicating support for reception of InfoFrames shall accept any of the InfoFrames defined here.

The second paragraph of Section 6.4 of CTA-861-I [A] is modified, as shown below:

> If the Source supports the transmission of the Auxiliary Video Information (AVI) InfoFrame and if it determines that the Sink is capable of receiving that information, it shall send the AVI InfoFrame to the Sink once per Video Field. The data from the AVI InfoFrame applies to the next full frame of video data.

The fifth paragraph of Section 6.4 of CTA-861-I [A] is modified, as shown below:

> If a Sink is capable of receiving YC_BC_R Pixel Data, then, as defined in Section 7.1, it is required to include the Version 3 CTA Extension in its EDID with at least one of the YC_BC_R chroma sampling format bits set. When a Sink's EDID indicates that it is capable of receiving YC_BC_R Pixel Data the Sink shall be capable of receiving AVI InfoFrames. If no AVI InfoFrame is sent from the Source, then, as defined in Section 5.1, the Sink is required to assume that all CE Video Formats are encoded in RGB color space using Limited Range levels.

The second paragraph before Table 14 of Section 6.4.1 of CTA-861-I [A] is modified, as shown below:

> Data Byte 2, bits M1 and M0, contain the Picture Aspect Ratio. When VIC = 0 and RID = 0, the M field may be used to signal a Picture Aspect Ratio. When M = 0 (M1 = 0, M0 = 0) and VIC = 0, and RID = 0, the Picture is considered to be formatted according to the Preferred Picture Aspect Ratio. If the VIC is not 0, or the RID is not 0, then the Sink shall ignore this field and instead use the Picture Aspect Ratio entry in Table 2 associated with the non-zero VIC, or the Picture Aspect Ratio entry in Table 11 associated with the non-zero RID, and the Source shall set M to 0, or to the matching Picture Aspect Ratio of the VIC or RID.

The last sentence of the third paragraph before Table 16 in Section 6.4.1 of CTA-861-I [A] is modified, as shown below:

> Table 17 gives recommended Bar placement for each value of PR in the form of Bar Data (see Section 6.4.4).

The heading of Table 16 in Section 6.4.1 of CTA-861-I [A] is modified, as shown below:

VIC	Video Description	Valid Pixel Repeat Values	PR Required

The heading of Table 17 in Section 6.4.1 of CTA-861-I [A] is modified, as shown below:

Unique Active Pixel Spacing (in Video Pixels)	Unique Active Pixels	Unique Content Pixels	PR	B	ELB	SRB

The first paragraph of Section 7.5.1 of CTA-861-I is modified, as shown below:

> A Video Data Block (VDB) lists Video Formats associated with a Video Identification Code (VIC) in Table 2 that are supported by the Sink. Formats enumerated in a VDB do not support YC_BC_R 4:2:0 sampling mode unless indicated in a YC_BC_R 4:2:0 Capability Map Data Block (see Section 7.5.11). VIC-associated Video Formats supported by the Sink using only YC_BC_R 4:2:0 sampling mode are not to be indicated using the Video Data Block. (See Section 7.5.10 on how to indicate such Video Formats.) Each supported Video Format is represented by a Short Video Descriptor (SVD) containing a VIC and, in the case of VICs 1 through 64, a Native Video Format indicator.

The fifth paragraph below Table 86 (now Table 85) of Section 7.5.6 of CTA-861-I is modified, as shown below:

> If the display declares that it can support both overscan and underscan for a Video Format category and the Source outputs that type, then the display shall either automatically overscan or underscan (in response to the AVI InfoFrame S field) or provide user options of selecting an overscan or an underscan mode. If operating in an automatic mode, the display shall overscan the incoming Picture if it receives AVI S = 1 and it shall underscan if it receives AVI S = 2. The Source shall always set the AVI S field correctly if that information is known by the Source. If the display receives no AVI InfoFrame or AVI S = 0, then the display should overscan CE Video Formats and underscan IT Video Formats by default but may provide a user-selectable alternative behavior.

The first paragraph of Section 7.5.10 of CTA-861-I is modified, as shown below:

> A YC_BC_R 4:2:0 Video Data Block (Y420VDB) lists Video Formats associated with a Video Identification Code (VIC) in Table 2 that are supported by the Sink only using YC_BC_R 4:2:0 sampling mode (i.e., not RGB, YC_BC_R 4:4:4, or YC_BC_R 4:2:2 sampling modes). Video formats represented as VICs that support RGB, YC_BC_R 4:4:4, or YC_BC_R 4:2:2 sampling modes, in addition to YC_BC_R 4:2:0 sampling mode, shall be listed in a regular Video Data Block (see Section 7.5.1) and marked as supporting YC_BC_R 4:2:0 sampling mode using a YC_BC_R 4:2:0 Capability Map Data Block (see Section 7.5.11).

The first paragraph of Section 7.5.17.1 of CTA-861-I [A] is modified, as shown below:

> The Type VII Video Timing Data Block (T7VTDB) is an expansion of the legacy 18-byte Detailed Timing Descriptor from the E-EDID v1.3 standard [9]. Whereas 18-byte DTDs are limited to 4095 pixels, 4095 lines, and a 655.35 MHz pixel clock, these 20-byte DTDs support up to 64k pixels and lines, and pixel rates up to 16 GHz.

The third and fourth paragraphs of Section 7.5.17.1 of CTA-861-I are modified, as shown below:

> Byte 3 contains the revision of the data structure and the variable-length indication of the descriptor. The revision supported by CTA-861 is 0b010; other revisions are unsupported. The DSC_PT field has reserved usage for VESA DisplayID; for use with CTA-861, it shall be cleared (=0). The T7_M field identifies the length of the descriptor beyond the original 20-byte length: the descriptor length = 20 + T7_M. The T7_M shall be 0b000; all other values are reserved.
>
> The T7Y420 field indicates support for YC_BC_R 4:2:0 sampling mode. (For Pixel Encoding support, see Section 7.5.17.4.)

The last paragraph of Section 7.5.17.1 of CTA-861-I is modified, as shown below:

> Any 20-byte DTDs described in T7VTDBs may also be used in the Video Format Preference Data Block. The T7VTDB shall not be used to advertise any video timing that can be exactly expressed within the defined ranges and precisions of the fields in an 18-byte DTD. The T7VTDB should not be used to advertise any video timing that can be exactly expressed using a T10VTDB.

The first paragraph of Section 7.5.17.2 of CTA-861-I is modified, as shown below:

> The Type VIII Video Timing Data Block (T8VTDB) is similar to the Video Data Block (VDB) in that it can contain a list of single-byte codes that represent timing formats from a given standard. The T8VTDB may also be used with two-byte codes. In DisplayID, the codes are sourced from

the VESA DMT standard, CTA-861, and HDMI 1.4b. The T8VTDB shall not be used with CTA-861 or HDMI 1.4b single-byte VICs as those codes have more-efficient locations in the EDID (the VDB and the HDMI VSDB, respectively). A future version of CTA-861 could potentially expand its VIC usage to two bytes for which the T8VTDB can be used.

The fourth paragraph of Section 7.5.17.2 of CTA-861-I is modified, as shown below:

Table 110 shows the structure of the T8VTDB within a CTA Extended Data Block using the Extended Tag Code of 35 (0x23). The payload for this Data Block is at minimum one code, and at maximum 29 one-byte codes or 14 two-byte codes.

The fifth paragraph of Section 7.5.17.2 of CTA-861-I is modified, as shown below:

Byte 3 contains revision and YC_BC_R 4:2:0 support identical to that of the T7VTDB. The revision supported by CTA-861 is 0b001; other revisions are unsupported. The T8Y420 indicates support for YC_BC_R 4:2:0 sampling mode. (For Pixel Encoding support, see Section 7.5.17.4.)

The first paragraph of Section 7.5.17.3 of CTA-861-I is modified, as shown below:

The Type X Video Timing Data Block (T10VTDB) is used to advertise IT timing formats in a more efficient manner than a DTD, as each entry requires six or seven bytes. The timing formats listed in the T10VTDB must conform to the VESA Coordinated Video Timing (CVT) formula, and may use standard- or reduced-blanking. Since CE timing formats in CTA-861 do not conform to the CVT standard, VIC timing formats cannot be advertised in the T10VTDB. Multiple T10VTDBs may exist in an EDID. The T10VTDB should not be used to advertise a video timing that can be expressed using a Standard Timing ID two-byte code; such a two-byte code can be used in the Standard Timings area of the Base Block (offsets 0x26 – 0x34) or in a T8VTDB if the Standard Timings area is full. Note that T10VTDBs cannot implement any of the Established Timings.

The third paragraph of Section 7.5.17.3 of CTA-861-I is modified, as shown below:

Byte 3 contains the revision of the data structure and the variable-length indication of the descriptor. The revision supported by CTA-861 is 0b000; other revisions are unsupported. The T10_M field identifies the length of the descriptor beyond the original 6-byte length: the descriptor length = 6 + T10_M. The T10_M shall be 0b000 or 0b001; all other values are reserved. All descriptors in a T10VTDB are of the same length as indicated by T10_M.

The paragraphs eight and nine of Section 7.5.17.3 of CTA-861-I are modified, as shown below:

> The T10Y420 field indicates support for YC_BC_R 4:2:0 sampling mode. (For Pixel Encoding support, see Section 7.5.17.4.)
>
> The CVT formula takes in a number of parameters: a Timing Formula enumeration for the algorithm; the active horizontal pixel and vertical line counts; the refresh rate in Hz; and a Boolean for either the refresh rate ratio or horizontal blank. When the Timing Formula is RB3, the parameters also include a Delta Horizontal Blank value; an Additional Vertical Blanking Time; and an Early Vsync Flag. The 3D_Support field is encoded as per Table 109. The Timing_Formula field is encoded as per Table 117. Sources shall ignore T10VTDB descriptors containing Timing_Formula values that are reserved or not supported.

Section 7.5.17.4, including title, of CTA-861-I is modified, as shown below:

> **7.5.17.4 Supported Pixel Encodings for Video Format Data Blocks and DisplayID Video Timings Data Blocks**
>
> Subject to any IDO constraints (e.g., interface clock limits), the following Pixel Encodings shall be supported by the formats in the VTDBs and VFDBs:
>
> - RGB is always supported.
> - YC_BC_R 4:4:4 and 4:2:2, are only supported if indicated in byte 3 bits 5..4 of the CTA Extension.
> - YC_BC_R 4:2:0 is only supported when the respective bit is set to one in the VTDB or VFDB.

The fifth paragraph of Section 7.5.19 of CTA-861-I is modified, as shown below:

> The Y420 field in Byte 2 indicates support for YC_BC_R 4:2:0 sampling mode. (For Pixel Encoding support, see Section 7.5.17.4.)

The first and second paragraphs of Section 7.5.19.1 of CTA-861-I, and Table 121 (now Table 120), are modified, as shown below:

> A Video Format Descriptor (VFD) indicates a single Video Resolution at one or more Frame Rates supported by the Sink. One or more VFDs are listed in a Video Format Data Block (VFDB) (see Section 7.5.19). Each VFD can represent Video Formats of a single resolution and aspect ratio, and multiple frame rates. Each VFD shall indicate at least one supported frame rate. The length of a VFD is indicated by the L_{vfd} field in the VFDB containing the VFD. The size of a VFD is (L_{vfd} + 1) bytes.

Table 120 - Video Format Descriptor

	Bit							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	Frame Rate		Resolution					
	BFR50	FR24	RID					
2	Frame Rate		Frame Rate Factor					
(If L_{vfd} >= 1)	BFR60	FR144	8X	6X	4X	2X	1X	0.5X
3							Frame Rate	
(If L_{vfd} >= 2)	F37=0	F36=0	F35=0	F34=0	F33=0	F32=0	F31=0	FR48
4								
(If L_{vfd} = 3)	F47=0	F46=0	F45=0	F44=0	F43=0	F42=0	F41=0	F40=0

VFDs are currently defined with a length of 1 to 3 bytes. Future versions of this Specification might define VFD lengths up to 4 bytes. If Byte 2 is not present in the VFD, flags 0.5X, 1X, and BFR60 shall be considered set, indicating support for 29.97 Hz, 30 Hz, 59.94 Hz, and 60 Hz, and flags 2X, 4X, 6X, 8X, and FR144 shall be considered cleared. If Byte 3 is not present in the VFD, flag FR48 shall be considered cleared. Byte 4 is reserved in this version of the Specification.

Table 122 (now Table 121) in Section 7.5.19.3 of CTA-861-I is modified, as shown below:

Table 121 - Frame Rate Support

		Base Frame Rate	
		50 Hz (BFR50 = 1)	**59.94 or 60 Hz (BFR60 = 1)**
Rate Factor	**0.5 (0.5X = 1)**	25 Hz	29.97 Hz or 30 Hz
	1.0 (1X = 1)	50 Hz	59.94 Hz or 60 Hz
	2.0 (2X = 1)	100 Hz	119.88 Hz or 120 Hz
	4.0 (4X = 1)	200 Hz	239.76 Hz or 240 Hz
	6.0 (6X = 1)	300 Hz	395.64 Hz or 360 Hz
	8.0 (8X = 1)	400 Hz	479.52 Hz or 480 Hz

The third paragraph of Section 4 of CTA-861-I is modified, as shown below:

A Video Timing with a vertical frequency that is an integer multiple of 6.00 Hz (e.g., 24.00 or 120.00 Hz) is considered to be the same as a Video Timing with the equivalent detailed timing information but where the vertical frequency is adjusted by a factor of 1000/1001 (e.g., 24/1.001 or 120/1.001). That is, they are considered two versions of the same Video Timing but with slightly different pixel clock frequencies. Therefore, a DTV that declares it is capable of displaying a Video Timing with a vertical frequency that is either an integer multiple of 6 Hz or an integer multiple of 6 Hz adjusted by a factor of 1000/1001 shall be capable of displaying both versions of the Video Timing. 300 Hz, declared in a VFD (see Section 7.5.19.1) with the BFR50 and 6X flags, is excepted from this requirement. Frame Rates that are multiples of

BFR60 are considered integer multiples of 6.00 Hz while those that are multiples of BFR50 are not. A Sink cannot declare 300 Hz with a BFR60 flag in a VFD, because no 5X flag exists.

A new definition is added to Section 2.2 of CTA-861-I, after Bars, as shown below:

Bars: Region of the display screen that is being driven or scanned at either zero luminance or at a uniform luminance; or regions of a Picture that are intended to be driven (e.g., matrix addressed) or scanned (e.g., cathode ray tube (CRT)) at either zero luminance or at a uniform luminance. In other words, it is the portion of the Picture that does not contain useful information (see Annex N).

Base Frame Rate: Frame Rate of a Video Format advertised in a VFD, where support for different multiples of the Base Frame Rate can be efficiently indicated using Frame Rate Factor flags.

Basic Audio: Uncompressed, two channel, digital audio. Exact parameters are determined by the interface specification used with CTA-861 (e.g., 2 channel IEC 60958-3 [12] L-PCM, 32, 44.1, and 48 kHz sampling rates, 16 bits/sample).

Two new abbreviations are added to Section 2.3 of CTA-861-I, after BDA and after VESA DTCDB, respectively, as shown below:

BDA Blu-ray Disc Association

BFR Base Frame Rate

CD Compact Disk

[...]

VESA DTCDB VESA Display Transfer Characteristic Data Block

VFD Video Format Descriptor

VFPDB Video Format Preference Data Block

7 PRODUCT ID

Entries [127] and [128] of Section 2.1.2.1 of CTA-861-I [A] are modified, as shown below:

127. *Unified Extensible Firmware Interface Forum* — PNP ID and ACPI ID Registry, https://uefi.org/PNP_ACPI_Registry.
128. *IEEE Registration Authority*, https://standards.ieee.org/products-programs/regauth/.

Table 64 - CTA Extended Tag Codes in Section 7.4 of CTA-861-I is modified, as shown below:

Table 64 - CTA Extended Tag Codes

CTA Extended Tag Code	Type of Data Block
0	Video Capability Data Block
1	Vendor-Specific Video Data Block
2	VESA Display Device Data Block [135]
3	Reserved for VESA Video Data Block
4	Reserved for HDMI Video Data Block [140]
5	Colorimetry Data Block
6	HDR Static Metadata Data Block
7	HDR Dynamic Metadata Data Block
8	Native Video Resolution Data Block
9…12	Reserved for video-related blocks
13	Video Format Preference Data Block
14	YC_BC_R 4:2:0 Video Data Block
15	YC_BC_R 4:2:0 Capability Map Data Block
16	Reserved for CTA Miscellaneous Audio Fields (MAF)
17	Vendor-Specific Audio Data Block
18	HDMI Audio Data Block [140]
19	Room Configuration Data Block
20	Speaker Location Data Block
21…31	Reserved for audio-related blocks
32	InfoFrame Data Block (includes one or more Short InfoFrame Descriptors)
33	Product Information Data Block
34	DisplayID Type VII Video Timing Data Block
35	DisplayID Type VIII Video Timing Data Block
36..41	Reserved
42	DisplayID Type X Video Timing Data Block
43…119	Reserved
120	HDMI Forum EDID Extension Override Data Block [140]
121	HDMI Forum Sink Capability Data Block [140]
122	HDMI Forum Source-Based Tone Mapping Data Block [140]
123…127	Reserved for HDMI
128…255	Reserved

A new Section 7.5.20, including a new Table 122, is added to CTA-861-I, and the remaining Tables are re-numbered accordingly. Refer to Table A for details.

7.5.20 Product Information Data Block (PIDB)

This Data Block contains an IEEE Company ID (CID) or Organizationally Unique Identifier (OUI) with 3 Bytes. Existing OUIs may be used, but new assignments should use CIDs. Refer to the IEEE Registration Authority [128] for details.

If the PNP ID for Manufacturer Name in Bytes 0x08 ~ 0x09 of Block 0 of the EDID is set to 'CID', this data block provides the manufacturer ID. The PNP ID for Manufacturer Name in Bytes

0x08 ~ 0x09 of Block 0 of the EDID shall be set to either 'CID', or a PNP ID indicating the same manufacturer as the CID or OUI in this data block.

Optionally, a Version can be indicated in Byte 6. If no Version is present, Length L shall be set to 4.

- If the Version is present and has a value of 0, a Product Name is indicated in the remaining bytes. The Product Name consists of up to 25 7-bit ASCII characters. If this Product Name is present, the Monitor Name Descriptor shall be ignored.

Future versions of this Specification might define other Version values, indicating other uses for the remaining payload bytes.

The PIDB for Version = 0, or no Version, is shown in Table 122.

Table 122 - Product Information Data Block (PIDB) for Version 0

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x07)			Length (L) = number of bytes following this byte				
2	CTA Extended Tag Code = 33 (0x21)							
3	IEEE CID or OUI third two hex digits							
4	IEEE CID or OUI second two hex digits							
5	IEEE CID or OUI first two hex digits							
6 (If L >= 5)	Version = 0 (optional)							
7 through L + 1	Product Name (Max 25 characters, optional)							

Table 123 in Section 7.6 of CTA-861-I is modified, as shown below:

Table 123 - Rule List for Data Blocks

Tags	Extended Tags	Types of Data Blocks	Multiple Instances	Concatenation
1, 2, 6	Not Applicable	VDB, ADB, VFDB	Allowed	Any
3	Not Applicable	VSDB	Allowed	Vendor-Defined
4, 5	Not Applicable	SADB, VESA DTCDB	Not Allowed	Not Applicable
7	0, 2, 5, 6, 8	VCDB, VESA DDDB, CDB, HDR SMDB, NVRDB	Not Allowed	Not Applicable
7	1	VSVDB	Allowed	Vendor-Defined
7	3	VESA VTBExt (Not a DB)	Reserved	Not Applicable
7	4	Reserved for HDMI VDB [a]	Reserved	Not Applicable
7	7, 14	HDR DMDB, Y420VDB	Allowed	Any
7	13, 15	VFPDB, Y420CMDB	Not Allowed	Not Applicable
7	16	CTA MAF	Reserved	Not Applicable
7	17	VSADB	Allowed	Vendor-Defined
7	18	HDMI ADB [a]	Not Allowed	Not Applicable
7	19	RCDB	Not Allowed	Not Applicable
7	20, 32	SLDB, IFDB	Allowed	Any
7	33	PIDB	Not Allowed	Not Applicable

7	34, 35, 42	T7VTDB, T8VTDB, T10VTDB	Allowed	Any
7	120	HDMI Forum EEODB [a]	Not Allowed	Not Applicable
7	121	HDMI Forum SCDB [a]	Not Allowed	Not Applicable
7	122	HDMI Forum SBTMDB [a]	Not Allowed	Not Applicable
7	123 - 127	Blocks reserved for HDMI	See latest HDMI Specification	
7	All others	Reserved / Undefined	Reserved	Not Applicable

Notes:
a. For definition, see HDMI Forum's latest HDMI Specification

The heading and first two paragraphs of Annex A.2.4 of CTA-861-I [A] are modified, as shown below:

A.2.4 Vendor / Product Identification

This section's example starts and ends with address locations 0x08 and 0x11. Byte allocation for each location is as follows:

0x08 ~ 0x09 are a two byte PNP ID for Manufacturer Name and should contain a valid identification number. Refer to Section 7.5.20 if the PNP ID is "CID". Data for these bytes is based upon compressed ASCII, for example: "CEA", the PNP ID of the Consumer Technology Association, is created by using five-bit codes, where "C" = 0b00011. "E" = 0b00101, and "A" = 0b00001. Table 126 illustrates the address location and sample data for Manufacturer's Name, which is "CEA". For information on how to obtain a PNP ID, see Unified Extensible Firmware Interface Forum — PNP ID AND ACPI ID REGISTRY [127].

8 CONTENT INDICATION

The fifth paragraph of Section 6.4.5 of CTA-861-I [A] is modified, as shown below:

The IT content bit in Data Byte 3 (ITC) indicates when Picture content is directly composed according to common IT practices or derived from a specific type of IT content. When the IT content bit of Data Byte 3 is set to 1 (ITC = 1), the content field (CN0, CN1) of Data Byte 5 is valid and downstream processors should process Pixel Data according to the definitions given in Table 33. When the IT content bit of Data Byte 3 is false (ITC = 0), the content field (CN0, CN1) of Data Byte 5 should be ignored by the Sink and should be set to 0 by the Source. The IT content bit is about the Picture content and should not be confused with IT vs CE video timings. The IT content bit can be used with both video timings.

A new paragraph, including new footnote 13, is inserted ahead of the sixth paragraph of Section 6.4.5 of CTA-861-I, and the sixths paragraph is modified, as shown below. The remaining footnotes are re-numbered accordingly:

> Some IDOs provides a different definition for the ITC and CN bits. Unless otherwise defined by an IDO, Sources should set ITC and CN to 0[13].
>
> Table 33 illustrates the meaning of Data Byte 5 Content Type bits CN1 and CN0. These bits may be used to signal delivery of IT content that is either classified as Graphics, Photo, Cinema, or Game.
>
> [13] The definitions of how Sinks behave for non-0 CN values are insufficiently detailed in CTA-861, leading to unpredictable behavior which makes the use of non-0 CN values impractical.

The first and second paragraphs of Section 7.5.5 of CTA-861-I are modified, as shown below:

> The Colorimetry Data Block (CDB) indicates support of specific extended colorimetry standards and IDO defined gamut-related metadata. Details regarding the contents of the CDB are provided in Table 82 and Table 83.
>
> Byte 3 and bits 4 through 7 of Byte 4 are allocated for Colorimetry data. The definitions of the colorimetry flags are shown in Table 83. Setting a colorimetry flag to one shall indicate that the Sink is capable of displaying Pictures encoded in that colorimetry. Bit 0 of Byte 4 (MD0) indicates support for IDO-defined gamut-related metadata.

Table 82 in Section 7.5.5 of CTA-861-I is modified, as shown below:

Table 82 - Colorimetry Data Block (CDB)

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x07)			Length of following data block (in bytes) (0x03)				
2	CTA Extended Tag Code (0x05)							
3	$BT2020_{RGB}$	$BT2020_{YCC}$	$BT2020_{cYCC}$	opRGB	$opYCC_{601}$	$sYCC_{601}$	$xvYCC_{709}$	$xvYCC_{601}$
4	$ST2113_{RGB}$[25]	IC_TC_P	sRGB	defaultRGB	F43=0	F42=0	F41=0	MD0

Table 84 - Data Byte 4 Colorimetry Metadata Support Flags in Section 7.5.5 of CTA-861-I is removed, and the remaining Tables are re-numbered accordingly. Refer to Table A for details.

9 3D AUDIO

The first paragraph in Section 6.6 of CTA-861-I [A] is modified, as shown below:

> The Audio InfoFrame contains information that allows for the format of the digital audio streams to be identified more quickly via out-of-band information and, for multi-channel Uncompressed Audio (which does not otherwise give such information), provides channel allocation information for the Sink's speakers. The Audio InfoFrame format is shown in Table 36, Table 37, and Table 38.

Table 42 in Section 6.6.2 of CTA-861-I [A] is modified, as shown below:

Table 42 - Speaker Placement

Label	Position Description	Speaker ID
FL	Front Left	0x00
FR	Front Right	0x01
FC	Front Center	0x02
LFE1	Low Frequency Effects 1	0x03
BL	Back Left	0x04
BR	Back Right	0x05
FLc	Front Left of Center	0x06
FRc	Front Right of Center	0x07
BC	Back Center	0x08
LFE2	Low Frequency Effects 2	0x09
SiL	Side Left	0x0A
SiR	Side Right	0x0B
TpFL	Top Front Left	0x0C
TpFR	Top Front Right	0x0D
TpFC	Top Front Center	0x0E
TpC	Top Center	0x0F
TpBL	Top Back Left	0x10
TpBR	Top Back Right	0x11
TpSiL	Top Side Left	0x12
TpSiR	Top Side Right	0x13
TpBC	Top Back Center	0x14
BtFC	Bottom Front Center	0x15
BtFL	Bottom Front Left	0x16
BtFR	Bottom Front Right	0x17
FLw	Front Left Wide	0x18
FRw	Front Right Wide	0x19
LS	Left Surround	0x1A
RS	Right Surround	0x1B
	reserved	0x1C
	reserved	0x1D
	reserved	0x1E

Label	Position Description	Speaker ID
	No named speaker indicated	0x1F

Rule 5 in Section 6.6.3 of CTA-861-I [A] is modified, as shown below:

5) If an odd number of channels is being presented, then an inactive channel shall occupy the second channel of the channel pair carrying the single channel. Inactive channels should be muted with all sample bits set to 0 (i.e., zero volume) and the values for V, C, U, and P for inactive channels should contain correct information, as defined in IEC 60958-3 [12].

The first paragraph of Section 6.4.4 of CTA-861-I [A], is modified, as shown below:

Channels being delivered by the method described here require an assigned Channel Index (see Table 105 in Section 7.5.16), therefore an SLD is required for each channel using this delivery method. A Source shall only deliver audio data by channel index when a Speaker Location Data Block is available for all channels being utilized by the Source.

The second paragraph of Section 7.5.2 of CTA-861-I [A], is modified, as shown below:

The Short Audio Descriptor shall conform to the formats given in Table 68 through Table 79 as a function of the Audio Format Code. Several types of audio may be supported, but each one shall be listed in its own Short Audio Descriptor with its designated code and the associated information. Multiple SADs for the same Audio Format Code may be present to indicate support for different configurations. The list of Audio Format Code values is given in Table 39 and Table 41.

A new paragraph is added before Table 80 in Section 7.5.3 of CTA-861-I [A], as shown below:

When delivering audio according to the Speaker Mask (Section 6.6.3), the contents of the SADB are not used.

Table 80, including footnote 23, in Section 7.5.3 of CTA-861-I [A] is modified, and a new footnote 24 is added, as shown below. The remaining footnotes are re-numbered accordingly:

Table 80 - Speaker Allocation Data Block (SADB)[22]

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x04)			Length of following data block (in bytes) (0x03)				
2	FLw/ FRw	F16=0[23]	FLc/FRc	BC	BL/BR	FC	LFE1	FL/FR
3	F27=0[24]	F26=0[24]	F25=0[24]	F24=0[24]	LS/RS	TpFC	TpC	TpFL/ TpFR
4	F37=0	F36=0	F35=0	F34=0	F33=0[25]	F32=0[24]	F31=0[24]	F30=0[24]

[23] Use of F16 for RLC/RRC has been deprecated. RLC/RRC are considered to be logically the same speaker positions as BL/BR. Legacy Sinks might use RLC/RRC instead of BL/BR; Sources shall route audio intended for RLC/RRC to BL/BR. See Table 37 and Table 43. Future use of F16 for other speaker positions is reserved.

[24] Use of F24 for LFE2, F25 for TpBC, F26 for SiL/SiR, F27 for TpSiL/TpSiR, F30 forTpBL/TpBR, F31 for BtFC, and F32 for BtFL/BtFR has been deprecated in the SADB. Use the RCDB (see Section 7.5.15) for these speakers.

The first paragraph of Section 7.5.15 of CTA-861-I [A], is modified, as shown below:

The Room Configuration Data Block (RCDB) provides a more complete speaker layout and room description than is possible with the Speaker Allocation Data Block. There are three progressively increasing levels of speaker location detail using the RCDB:

1) Using only the RCDB Speaker Presence Mask (SPM), allowing a selection of speakers and speaker pairs supported by the system. See Section 6.6.3 (Delivery According to the Speaker Mask) on how to deliver audio to specific speakers. The SADB is not used when delivering according to the Speaker Mask.
2) Adding the Speaker Location Data Block (SLDB) with an associated Speaker Location Descriptor (SLD) for each speaker using the speaker labels defined in Table 42. See Section 6.6.3 (Delivery According to the Speaker Mask) on how to deliver audio to specific speakers.
3) Adding spatial coordinates to one or more of the SLDs using the room coordinate system defined in Section 7.5.16.1, permitting a Source to use these spatial coordinates when using Delivery by Channel Index (see Section 6.6.4).

A new paragraph is added before the fourth paragraph of Section 7.5.15 of CTA-861-I [A], and that fourth paragraph (now fifth paragraph) is modified, as shown below:

> The number of active speakers indicated in the RCDB Speaker Count or with the SPM in the RCDB shall be less than or equal to Max Number of Channels in MC4:MC0 of at least one SAD with Audio Coding Extension Type 0x0D (L-PCM 3D Audio, see Table 77 in Section 7.5.2).
>
> Information in the RCDB SPM can differ from the speaker definitions in the Speaker Allocation Data Block. Generally, it is preferred that the capabilities advertised in the RCDB SPM are equal or exceed those in the SADB. The Source may rely upon either the SADB, the RCDB SPM, or the SLDB(s) if present in the Sink.

The first through fifth paragraphs below Table 104, now Table 103, and footnote 28 (now 29) of that table in Section 7.5.15 of CTA-861-I [A], are modified, as shown below:

> [29] Use of F46 for RLC/RRC has been deprecated. RLC/RRC are considered to be logically the same speaker positions as BL/BR. Legacy Sinks might use RLC/RRC instead of BL/BR; Sources shall route audio intended for RLC/RRC to BL/BR. See Table 37 and Table 43. Future use of F46 for other speaker positions is reserved.

> If Display = 1, then room coordinates locating the display are available, and Xmax, Ymax, Zmax, DisplayX, DisplayY, and DisplayZ shall be present. If Display = 0, then the display location is not indicated.
>
> If Speaker = 1, then Speaker Count shall be valid. If Speaker = 0, then Speaker Count shall not be used to determine the number of PCM audio channels.
>
> If SLD = 1, then Speaker Location Descriptors shall be present for every speaker defined by the system and Speaker shall be set to 1. If SLD = 0, then Speaker Location Descriptors are not available, Speaker shall be cleared to 0, and only the SPM fields shall be read to describe the speaker layout.
>
> The Speaker and SLD flags shall be set to the same value. When SLD = 1, the 5 bit unsigned integer value Speaker Count represents the number of speakers indicated in SLDs. The value is set to one less than the total number of SLDs with Active = 1, representing from 1 to 32 active channels. When SLD = 0, Speaker Count shall be set to 0.
>
> The speaker designations used in the Speaker Presence Mask (SPM1 – SPM3) are the same as are used in the Audio InfoFrame (see Figure 6 and Table 42). The placement of the SPM1 – SPM3 fields are identical to Bytes 2, 3, and 4 in the SADB defined in Section 7.5.3. The flags in these 3 bytes indicate the presence of speakers by name. Speakers present shall be indicated by setting the respective flag to 1 and the remaining flags shall be set to 0.

The paragraph above Table 106, now Table 105, in Section 7.5.16 of CTA-861-I [A], is modified, as shown below:

> The Speaker Location Descriptor, shown in Table 105, is a data structure for providing specific information about channels present. The Speaker Location Descriptor shall only be used when SLD is set to 1 and Speaker Count has been configured in the RCDB (see Section 7.5.15). It describes one PCM channel with a Channel Index, a Speaker ID (See Table 42), and optionally, speaker coordinates. It occupies 2 or 5 bytes, depending on the flags in the first byte.

Table 106 in Section 7.5.16 of CTA-861-I [A], now Table 105, is modified, as shown below:

Table 105 - Speaker Location Descriptor

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Channel	F17=0	COORD	Active	Channel Index (0 to 31)				
ID	F27=0	F26=0	F35=0	Speaker ID (0x00 to 0x1B) (from Table 42) or 0x1F (if none indicated)				
COORD1	X							
COORD2	Y							
COORD3	Z							

Section 7.5.16, from the fourth paragraph through the end, of CTA-861-I [A] is modified, as shown below:

> Channel Index is the index of the audio channel where the audio for the speaker is to be transmitted. See also Section 6.6.4 and Table 42.
>
> The Sink orders the channels by the Channel Index for transmission from the Source as indicated in Section 6.6.4.
>
> Speaker ID indicates the named location of the speaker as per Table 42, or if the location does not correspond to a named location, it is set to 0x1F.
>
> If Active = 1, and a Source transmits audio with an accompanying Audio InfoFrame with the corresponding CID Mask bit set to 1, then the channel shall be rendered by the Sink. If Active = 0, then the Source shall not transmit audio based on the contents of that SLD.
>
> A Sink may set COORD = 1 in an SLD, regardless of the indicated Channel ID. If COORD = 1, the fields COORD1, COORD2, and COORD3 shall be present in the Speaker Location Descriptor (SLD), and X, Y, and Z in the SLD and Xmax, Ymax, and Zmax in the RCDB shall be valid. Sinks should configure the Speaker ID to indicate the named location of the speaker most closely representing the position indicated by the coordinates.
>
> If COORD = 0, then COORD1, COORD2, and COORD3 fields shall not be present in the SLD and Speaker ID shall identify the corresponding named location of the speaker being defined (see

Table 42). In this case, Speaker ID shall not be 0x1F and the Speaker ID shall be unique for SLDs with COORD = 0.

COORD1 is a Coordinate Value as defined in Table 106 of Section 7.5.16.1. X * Xmax indicates the position of this speaker on the x-axis.

COORD2 is a Coordinate Value as defined in Table 106 of Section 7.5.16.1. Y * Ymax indicates the position of this speaker on the y-axis.

COORD3 is a Coordinate Value as defined in Table 106 of Section 7.5.16.1. Z * Zmax indicates the position of this speaker on the z-axis.

The total number of Active flags set to 1 shall match the Speaker Count in the Room Configuration Data Block. At most one SLD carrying a given Channel Index shall have the Active flag set to 1, any other SLD with a duplicated Channel Index shall set the Active flag to 0.

A Sink shall not set multiple SLDs with the same Speaker ID, COORD=0, and Active=1. A Sink may set multiple SLDs with the same Speaker ID, COORD=1, and Active=1; and a Sink may also duplicate that Speaker ID in a single SLD with COORD=0 and Active=1. When multiple SLDs with the same Speaker ID, COORD=1, and Active=1 are present in the SLDB(s), then the Sink shall not set the location (X, Y, Z) identically in any such SLD.

A discussion of room coordinates with several examples can be found in Annex V.

The Example in Section 6.6.3 of CTA-861-I [A] is modified, as shown below, to correct the surround speaker indication:

Example:

A source is sending the following channels:

FL, FR, LFE1, FC, LS, RS, BC, TpFL, TpFR, LFE2

The order of the channel pairs shall be as follows:

FL/FR,

LFE1/FC,

LS/RS,

BC/LFE2,

TpFL/TpFR

Example 1 in Annex V.1 of CTA-861-I [A] is modified, and a new paragraph added ahead of it, as shown below:

In the following examples, active speaker and channel indications are shown in bold, for emphasis.

Example 1: The Sink is a 5.1 speaker system, with three speakers in the front, two surrounds, and one subwoofer.

The EDID for such a system does not need to use a RCDB, and would carry an SADB as shown in Table 190:

Table 190 - Example 1 SADB indicating a 5.1 Speaker System

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x04)			Length of following data block (in bytes) (0x03)				
2	FLw/ FRw=0	F16=0	FLc/FRc=0	BC=0	BL/BR=0	**FC=1**	**LFE1=1**	**FL/FR=1**
3	F27=0	F26=0	F25=0	F24=0	**LS/RS=1**	TpFC=0	TpC=0	TpFL/ TpFR=0
4	F37=0	F36=0	F35=0	F34=0	F33=0	F32=0	F31=0	F30=0

The Source, if feeding all speakers with six L-PCM channels, would send the following three channel pairs, in the indicated order:

FL/FR, LFE1/FC, LS/RS

With that, it would carry an Audio InfoFrame as shown in Table 191, with CA = 0x0B (see Table 43) indicating FL, FR, LFE1, FC, LS, and RS. Note that the flags in Bytes 1, 2, and 3 are set for an example using an IEC 60958 audio format, which carries information making these flags redundant.

Table 191 - Example 1 Audio InfoFrame for 5.1 Speakers

InfoFrame Type Code	InfoFrame Type = 0x04							
InfoFrame Version Number	Version = 0x01							
Length of Audio InfoFrame	Length of Audio InfoFrame = 10							
Data Byte 1	CT3=0	CT2=0	CT1=0	CT0=0	F13=0	CC2=1	CC1=0	CC0=1
Data Byte 2	F27=0	F26=0	F25=0	SF2=0	SF1=0	SF0=0	SS1=0	SS0=0
Data Byte 3	F37=0	F36=0	F35=0	CXT4=0	CXT3=0	CXT2=0	CXT1=0	CXT0=0
Data Byte 4	**CA7=0**	**CA6=0**	**CA5=0**	**CA4=0**	**CA3=1**	**CA2=0**	**CA1=1**	**CA0=1**
Data Byte 5	DM_INH	LSV3	LSV2	LSV1	LSV0	F52=0	LFEPBL1	LFEPBL0
Data Byte 6	F67=0	F66=0	F65=0	F64=0	F63=0	F62=0	F61=0	F60=0
Data Byte 7	F77=0	F76=0	F75=0	F74=0	F73=0	F72=0	F71=0	F70=0
Data Byte 8	F87=0	F86=0	F85=0	F84=0	F83=0	F82=0	F81=0	F80=0
Data Byte 9	F97=0	F96=0	F95=0	F94=0	F93=0	F92=0	F91=0	F90=0
Data Byte 10	F107=0	F106=0	F105=0	F104=0	F103=0	F102=0	F101=0	F100=0

Example 2 in Annex V.1 of CTA-861-I [A] is modified, as shown below:

Example 2: The Sink is a 7.1.4 speaker system, with three speakers in the front, two side surrounds and two surrounds in the rear, four speakers on the ceiling, and one subwoofer. However, in this example, the Sink has a processing limitation, allowing for a maximum of 10 of the 12 channels at a time.

The SADB (shown in Table 192) would be used to carry a legacy speaker configuration, for which the CA values defined in Table 43 can be used by the Source. In this example, the Sink indicates 7.1.2 speakers, but only up to 8 of those 10 channels can be indicated with CA values. The Source chooses CA values, commonly matching the content, e.g., CA=0x13 for 7.1 speakers, or CA=0x2F for 5.1.2 speakers.

To allow the Source to indicate a set of audio signals for all four of the ceiling speakers in an Audio InfoFrame with CA=0xFE, an RCDB (shown in Table 193) is included in the EDID. Due to this Sink's limitation to 10 channels only 5 of the 7 ear level speakers are indicated, reflecting its maximum processing capacity.

Table 192 - Example 2 SADB indicating 7.1.2 for a 7.1.4 Speaker System

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x04)			Length of following data block (in bytes) = 3 (0x03)				
2	FLw/ FRw=0	F16=0	FLc/FRc=0	BC=0	**BL/BR=1**	**FC=1**	**LFE1=1**	**FL/FR=1**
3	F27=0	F26=0	F25=0	F24=0	**LS/RS=1**	TpFC=0	TpC=0	**TpFL/ TpFR=1**
4	F37=0	F36=0	F35=0	F34=0	F33=0	F32=0	F31=0	F30=0

Table 193 - Example 2 RCDB indicating 5.1.4 for a 7.1.4 Speaker System

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x07)			Length of following block payload (bytes) = 5 (0x05)				
2	CTA Extended Tag Code (0x13)							
3	Display=0	Speaker=0	SLD=0	Speaker Count = 0 (0x00)				
SPM1	FLw/FRw=0	F46=0	FLc/FRc=0	BC=0	BL/BR=0	**FC=1**	**LFE1=1**	**FL/FR=1**
SPM2	TpSiL/TpSiR=0	SiL/SiR=0	TpBC=0	LFE2=0	**LS/RS=1**	TpFC=0	TpC=0	**TpFL/TpFR=1**
SPM3	F67=0	F66=0	F65=0	F64=0	F63=0	BtFL/BtFR=0	BtFC=0	**TpBL/TpBR=1**

The Source, if feeding all RCDB-indicated speakers with ten L-PCM channels, would send the following five channel pairs, in the indicated order:

FL/FR, LFE1/FC, TpFL/TpFR, LS/RS, TpBL/TpBR

With that, it would carry an Audio InfoFrame as shown in Table 194:

Table 194 - Example 2 Audio InfoFrame for 5.1.4 Speakers

InfoFrame Type Code	InfoFrame Type = 0x04							
InfoFrame Version Number	Version = 0x01							
Length of Audio InfoFrame	Length of Audio InfoFrame = 10							
Data Byte 1	CT3=0	CT2=0	CT1=0	CT0=0	F13=0	CC2=1	CC1=1	CC0=1
Data Byte 2	F27=0	F26=0	F25=0	SF2=0	SF1=0	SF0=0	SS1=0	SS0=0
Data Byte 3	F37=0	F36=0	F35=0	CXT4=0	CXT3=0	CXT2=0	CXT1=0	CXT0=0
Data Byte 4	1	1	1	1	1	1	1	0
Data Byte 5	DM_INH	LSV3	LSV2	LSV1	LSV0	F52=0	LFEPBL1	LFEPBL0
Data Byte 6	FLw/FRw=0	F66=0	FLc/FRc=0	BC=0	BL/BR=0	**FC=1**	**LFE1=1**	**FL/FR=1**
Data Byte 7	TpSiL/ TpSiR=0	SiL/SiR=0	TpBC=0	LFE2=0	**LS/RS=1**	TpFC=0	TpC=0	**TpFL/ TpFR=1**
Data Byte 8	F87=0	F86=0	F85=0	F84=0	F83=0	BtFL/ BtFR=0	BtFC=0	**TpBL/ TpBR=1**
Data Byte 9	F97=0	F96=0	F95=0	F94=0	F93=0	F92=0	F91=0	F90=0
Data Byte 10	F107=0	F106=0	F105=0	F104=0	F103=0	F102=0	F101=0	F100=0

Example 3 in Annex V.2 of CTA-861-I [A] is modified, as shown below:

Example 3: The Sink is a 5.1 speaker system with a single added Front Left Wide speaker: four speakers in the front (FLw, FL, FC, FR), two surrounds in the rear (LS, RS), and one subwoofer (LFE).

The EDID for such a system needs to use a RCDB. The SADB, shown in Table 195, would still be present to provide legacy compatibility, and would be the same as for a 5.1 system, not representing the additional FLw speaker since only complete pairs can be indicated:

Table 195 - Example 3 SADB indicating 5.1 for a 5.1 + FLw Speaker System

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x04)			Length of following data block (in bytes) = 3 (0x03)				
2	FLw/ FRw=0	F16=0	FLc/FRc=0	BC=0	BL/BR=0	**FC=1**	**LFE1=1**	**FL/FR=1**
3	F27=0	F26=0	F25=0	F24=0	**LS/RS=1**	TpFC=0	TpC=0	TpFL/ TpFR=0
4	F37=0	F36=0	F35=0	F34=0	F33=0	F32=0	F31=0	F30=0

The RCDB (shown in Table 196), however, would be able to represent that extra Speaker, by including SLDs for each speaker in a SLDB (shown in Table 197). The SPM flags of the RCDB are the same as in the SADB, indicating 5.1 speakers.

Table 196 - Example 3 RCDB with SPM indicating 5.1 for a 5.1 + FLw Speaker System

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x07)			Length of following block payload (bytes) = 5 (0x05)				
2	CTA Extended Tag Code (0x13)							
3	Display=0	Speaker=1	SLD=1	Speaker Count = 7 (0x06)				
SPM1	FLw/FRw=0	F46=0	FLc/FRc=0	BC=0	BL/BR=0	**FC=1**	**LFE1=1**	**FL/FR=1**
SPM2	TpSiL/TpSiR=0	SiL/SiR=0	TpBC=0	LFE2=0	**LS/RS=1**	TpFC=0	TpC=0	TpFL/TpFR=0
SPM3	F67=0	F66=0	F65=0	F64=0	F63=0	BtFL/BtFR=0	BtFC=0	TpBL/TpBR=0

Table 197 - Example 3 SLDB indicating a 5.1 + FLw Speaker System

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x07)			Length of following block payload (bytes) = 15 (0x0F)				
2	CTA Extended Tag Code (0x14)							
3	F17=0	COORD=0	Active=1	Channel Index (0 to 31) = 0x00				
4	F27=0	F26=0	F25=0	Speaker ID = 0x00 (**FL**)				
5	F17=0	COORD=0	Active=1	Channel Index (0 to 31) = 0x01				
6	F27=0	F26=0	F25=0	Speaker ID = 0x01 (**FR**)				
7	F17=0	COORD=0	Active=1	Channel Index (0 to 31) = 0x02				
8	F27=0	F26=0	F25=0	Speaker ID = 0x03 (**LFE1**)				
9	F17=0	COORD=0	Active=1	Channel Index (0 to 31) = 0x03				
10	F27=0	F26=0	F25=0	Speaker ID = 0x02 (**FC**)				
11	F17=0	COORD=0	Active=1	Channel Index (0 to 31) = 0x04				
12	F27=0	F26=0	F25=0	Speaker ID = 0x1A (**LS**)				
13	F17=0	COORD=0	Active=1	Channel Index (0 to 31) = 0x05				
14	F27=0	F26=0	F25=0	Speaker ID = 0x1B (**RS**)				
15	F17=0	COORD=0	Active=1	Channel Index (0 to 31) = 0x06				
16	F27=0	F26=0	F25=0	Speaker ID = 0x18 (**FLw**)				

The Source, if feeding all speakers with seven L-PCM channels, would send the following four channel pairs:

FL/FR, LFE1/FC, LS/RS, FLw/NULL

The last pair carries the audio signal for the FLw speaker, and silence (i.e., all L-PCM sample values are at one constant value) in the second, inactive channel. With that, it would carry an Audio InfoFrame as shown in Table 198:

Table 198 - Example 3 Audio InfoFrame for 5.1 + FLw Speakers

InfoFrame Type Code	InfoFrame Type = 0x04							
InfoFrame Version Number	Version = 0x01							
Length of Audio InfoFrame	Length of Audio InfoFrame = 10							
Data Byte 1	CT3=0	CT2=0	CT1=0	CT0=0	F13=0	CC2=1	CC1=1	CC0=0
Data Byte 2	F27=0	F26=0	F25=0	SF2=0	SF1=0	SF0=0	SS1=0	SS0=0
Data Byte 3	F37=0	F36=0	F35=0	CXT4=0	CXT3=0	CXT2=0	CXT1=0	CXT0=0
Data Byte 4	1	1	1	1	1	1	1	1
Data Byte 5	DM_INH	LSV3	LSV2	LSV1	LSV0	F52=0	LFEPBL1	LFEPBL0
Data Byte 6	CID07=0	**CID06=1**	**CID05=1**	**CID04=1**	**CID03=1**	**CID02=1**	**CID01=1**	**CID00=1**
Data Byte 7	CID15=0	CID14=0	CID13=0	CID12=0	CID11=0	CID10=0	CID09=0	CID08=0
Data Byte 8	CID23=0	CID22=0	CID21=0	CID20=0	CID19=0	CID18=0	CID17=0	CID16=0
Data Byte 9	CID31=0	CID30=0	CID29=0	CID28=0	CID27=0	CID26=0	CID25=0	CID24=0
Data Byte 10	F107=0	F106=0	F105=0	F104=0	F103=0	F102=0	F101=0	F100=0

Examples 4 and 5 in Annex V.3 of CTA-861-I [A] are modified, as shown below. A new Table 207 is added, and the subsequent Tables are re-numbered accordingly. Refer to Table A for details.

Example 4: The User has three speakers at arbitrary positions in the room, at the ceiling towards the left front (-0.8,0.4,1.0), at the bottom right towards the front (1.0,0.7,-1.0), and slightly left and below the center of the rear wall (-0.2,-1.0,-0.3). These locations do not correlate with any of the named speakers (see Table 42 and Table 199).

The EDID for such a system needs to use an RCDB. An SADB, shown in Table 200, would still be present to provide legacy compatibility, and could in this case present a 5.1.2 configuration to allow audio with height information to be sent to the Sink, which can be mapped by the Sink onto the available speakers:

Table 200 - Example 4 SADB indicating 5.1.2 for a Three Arbitrary Speaker System

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x04)			Length of following data block (in bytes) = 3 (0x03)				
2	FLw/FRw=0	F16=0	FLc/FRc=0	BC=0	BL/BR=0	**FC=1**	**LFE1=1**	**FL/FR=1**
3	F27=0	F26=0	F25=0	F24=0	**LS/RS=1**	TpFC=0	TpC=0	**TpFL/ TpFR=1**
4	F37=0	F36=0	F35=0	F34=0	F33=0	F32=0	F31=0	F30=0

The RCDB (shown in Table 201), however, would be able to represent the precise speaker locations, and allow a Source device to map any audio to those speakers, and optionally provide additional information about the physical room size (In this example, 8.2 m by 5.6 m by 3.4 m) and display position (center of display at 50 cm above the center of the front wall).

The SPM is a copy of the SADB but is configured to indicate the ceiling speakers of a 5.1.2 configuration with TpSiL/TpSiR instead of TpFL/TpFR. TpSiL/TpSiR is only available in the SPM, and is not available in the SADB. No additional flags are set in the SPM since none of them match the information present in the SLDs with COORD=1. The three speakers with COORD=1 are represented by the SLDs in a SLDB, as shown in Table 202. For the values of Xmax, Ymax, and Zmax given in the RCDB, the location (in meters), of the Channel Index 0 speaker is at -3.27 x 1.14 x 1.7, the Channel Index 1 speaker is at 4.1 x 1.97 x -1.7, and the Channel Index 2 speaker is at -0.83 x -2.8 x -0.5:

Table 201 - Example 4 RCDB with SPM indicating 5.1.4 for a Three Arbitrary Speaker System

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x07)			Length of following block payload (bytes) = 11 (0x0B)				
2	CTA Extended Tag Code (0x13)							
3	Display=1	Speaker=1	SLD=1	Speaker Count = 3 (0x02)				
SPM1	FLw/FRw=0	F46=0	FLc/FRc=0	BC=0	BL/BR=0	**FC=1**	**LFE1=1**	**FL/FR=1**
SPM2	**TpSiL/TpSiR=1**	SiL/SiR-0	TpBC=0	LFE2=0	**LS/RS=1**	TpFC=0	TpC=0	TpFL/TpFR=0
SPM3	F67=0	F66=0	F65=0	F64=0	F63=0	BtFL/BtFR=0	BtFC=0	TpBL/TpBR=0
MAX1	Xmax = 8.2 m / 2 = 41 dm (0x29)							
MAX2	Ymax = 5.6 m / 2 = 28 dm (0x1C)							
MAX3	Zmax = 3.4 m / 2 = 17 dm (0x11)							
DISP1	DisplayX = at the center: 0.0 (0x00)							
DISP2	DisplayY = at front wall: 1.0 (0x40)							
DISP3	DisplayZ = 5 dm / 17 dm: 0.29 (0x13)							

Table 202 - Example 4 SLDB indicating Three Arbitrary Speakers

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x07)			Length of following block payload (bytes) = 16 (0x10)				
2	CTA Extended Tag Code (0x14)							
3	F17=0	COORD=1	Active=1	Channel Index (0 to 31) = **0x00**				
4	F27=0	F26=0	F25=0	Speaker ID = 31 (0x1F)				
5	X = -32.8 dm: **-0.8** (0xCD)							
6	Y = 11.2 dm: **0.4** (0x1A)							
7	Z = 17.0 dm: **1.0** (0x40)							
8	F17=0	COORD=1	Active=1	Channel Index (0 to 31) = **0x01**				
9	F27=0	F26=0	F25=0	Speaker ID = 31 (0x1F)				
10	X = 41.0 dm: **1.0** (0x40)							
11	Y = 19.6 dm: **0.7** (0x2D)							
12	Z = -17.0 dm: **-1.0** (0xC0)							
13	F17=0	COORD=1	Active=1	Channel Index (0 to 31) = **0x02**				
14	F27=0	F26=0	F25=0	Speaker ID = 31 (0x1F)				
15	X = -8.2 dm: **-0.2** (0xF3)							
16	Y = -20.8 dm: **-1.0** (0xC0)							
17	Z = -5.1 dm: **-0.3** (0xED)							

The Source, if feeding only the speakers with COORD=1 with three L-PCM channels, would send two channel pairs, with the first three channels in order of the Channel Index in the SLDs, and the fourth, inactive channel silent.

With that, it would carry an Audio InfoFrame as shown in Table 203:

Table 203 - Example 4 Audio InfoFrame for Three Arbitrary Speakers

InfoFrame Type Code	InfoFrame Type = 0x04							
InfoFrame Version Number	Version = 0x01							
Length of Audio InfoFrame	Length of Audio InfoFrame = 10							
Data Byte 1	CT3=0	CT2=0	CT1=0	CT0=0	F13=0	CC2=0	CC1=1	CC0=0
Data Byte 2	F27=0	F26=0	F25=0	SF2=0	SF1=0	SF0=0	SS1=0	SS0=0
Data Byte 3	F37=0	F36=0	F35=0	CXT4=0	CXT3=0	CXT2=0	CXT1=0	CXT0=0
Data Byte 4	1	1	1	1	1	1	1	1
Data Byte 5	DM_INH	LSV3	LSV2	LSV1	LSV0	F52=0	LFEPBL1	LFEPBL0
Data Byte 6	CID07=0	CID06=0	CID05=0	CID04=0	CID03=0	**CID02=1**	**CID01=1**	**CID00=1**
Data Byte 7	CID15=0	CID14=0	CID13=0	CID12=0	CID11=0	CID10=0	CID09=0	CID08=0
Data Byte 8	CID23=0	CID22=0	CID21=0	CID20=0	CID19=0	CID18=0	CID17=0	CID16=0
Data Byte 9	CID31=0	CID30=0	CID29=0	CID28=0	CID27=0	CID26=0	CID25=0	CID24=0
Data Byte 10	F107=0	F106=0	F105=0	F104=0	F103=0	F102=0	F101=0	F100=0

Example 5: The User has a set of standard Front Left, Center, and Right speakers, and a Subwoofer, also referred to as a "3.1" configuration. However, in this example, the Center speaker is implemented in a phantom speaker configuration, with one speaker placed above the Display, and a second speaker below the Display. In conjunction, those two speakers sound the same as one speaker in the center of the Display, due to psychoacoustic principles. Providing these locations of the Center speakers allows a Source to add height information to the audio signals for those speakers. The upper Center speaker is placed at (0.0, 1.0, 0.79), which is 0.5 above the center of the Display at (0.0, 1.0, 0.29), and the lower Center speaker is at (0.0, 1.0, -0.21), 0.5 below the center of the Display.

In addition to that, four surround speakers at arbitrary spatial positions are also present. Here, they are placed around the room to complement the front speakers like surround and ceiling speakers, but not adhering close enough to any default spatial positions to allow the use of named speaker locations. Two are in the rear, at (-0.75, -0.33, 0.20) and (0.88, -0.60, -0.10), and the other two are placed high up towards the front, at (-0.60, 0.33, 0.95) and (0.80, 0.50, 0.90).

The EDID for such a system needs to use an RCDB. An SADB, shown in Table 204, would still be present to provide legacy compatibility, and could in this case present a 5.1.2 configuration to feed the three speakers in the front and the subwoofer, while also allowing for some surround and height audio that the Sink can map to the extra speakers or add to the front speakers:

Table 204 - Example 5 SADB indicating 5.1.2 for a 3.1 with dual FC + Four Arbitrary Speaker System

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x04)			Length of following data block (in bytes) = 3 (0x03)				
2	FLw/ FRw=0	F16=0	FLc/FRc=0	BC=0	BL/BR=0	**FC=1**	**LFE1=1**	**FL/FR=1**
3	F27=0	F26=0	F25=0	F24=0	**LS/RS=1**	TpFC=0	TpC=0	**TpFL/ TpFR=1**
4	F37=0	F36=0	F35=0	F34=0	F33=0	F32=0	F31=0	F30=0

The RCDB (shown in Table 205), however, would be able to represent the standard Front Left, Front Right, and LFE1 speakers, the non-standard locations of the two Center speakers, and the precise locations of the four extra speakers. This example uses the same physical room size (8.2 m by 5.6 m by 3.4 m) and display position (center of display at 50 cm above the center of the front wall) as the previous example. All nine speakers are defined as SLDs, carried in two SLDBs, since a single SLDB would exceed 32 Bytes. They are shown in Table 206 and Table 207.

The SPM can provide a spatial configuration, up the processing capabilities of the Sink, for Sources that are not able to render to spatial coordinates. In this example, an extensive 9.1.6 configuration is shown, which would require support for 16 L-PCM channels.

For the values of Xmax, Ymax, and Zmax given in the RCDB, the location (in meters), of the speakers with COORD=1 are:

- Channel Index 3 (Front Center) speaker at 0.0 x 2.8 x 1.35
- Channel Index 4 (Front Center) speaker at 0.0 x 2.8 x -0.35
- Channel Index 5 speaker at -3.08 x -0.92 x -0.35
- Channel Index 6 speaker at 3.59 x -1.66 x -0.16
- Channel Index 7 speaker at -2.43 x 0.92 x -1.62
- Channel Index 8 speaker at 3.27 x 1.4 x 1.54

Table 205 - Example 5 RCDB with SPM indicating 9.1.6 for a 3.1 with dual FC + Four Arbitrary Speaker System

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x07)			Length of following block payload (bytes) = 11 (0x0B)				
2	CTA Extended Tag Code (0x13)							
3	Display=1	Speaker=1	SLD=1	Speaker Count = 9 (0x08)				
SPM1	**FLw/FRw=1**	F46=0	FLc/FRc=0	BC=0	**BL/BR=1**	**FC=1**	**LFE1=1**	**FL/FR=1**
SPM2	**TpSiL/TpSiR=1**	SiL/SiR-0	TpBC=0	LFE2=0	**LS/RS=1**	TpFC=0	TpC=0	**TpFL/TpFR=1**
SPM3	F67=0	F66=0	F65=0	F64=0	F63=0	BtFL/BtFR=0	BtFC=0	**TpBL/TpBR=1**
MAX1	Xmax = 8.2 m / 2 = 41 dm (0x29)							
MAX2	Ymax = 5.6 m / 2 = 28 dm (0x1C)							
MAX3	Zmax = 3.4 m / 2 = 17 dm (0x11)							
DISP1	DisplayX = at the center: 0.0 (0x00)							
DISP2	DisplayY = at front wall: 1.0 (0x40)							
DISP3	DisplayZ = 5 dm / 17 dm: 0.29 (0x13)							

Table 206 - Example 5, First SLDB indicating 3.1 with dual FC + Four Arbitrary Speakers

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x07)			Length of following block payload (bytes) = 27				
2	CTA Extended Tag Code (0x14)							
3	F17=0	COORD=0	Active=1	Channel Index (0 to 31) = **0x00**				
4	F27=0	F26=0	F25=0	Speaker ID = 0x00 (**FL**)				
5	F17=0	COORD=0	Active=1	Channel Index (0 to 31) = **0x01**				
6	F27=0	F26=0	F25=0	Speaker ID = 0x01 (**FR**)				
7	F17=0	COORD=0	Active=1	Channel Index (0 to 31) = **0x02**				
8	F27=0	F26=0	F25=0	Speaker ID = 0x03 (**LFE1**)				
9	F17=0	COORD=1	Active=1	Channel Index (0 to 31) = **0x03**				
10	F27=0	F26=0	F25=0	Speaker ID = 0x02 (**FC**)				
11	X = 0 dm: **0.0** (0x00)							
12	Y = 28.0 dm: **1.0** (0x40)							
13	Z = 5 + 8.5 = 13.5 dm: **0.79** (0x33)							
14	F17=0	COORD=1	Active=1	Channel Index (0 to 31) = **0x04**				
15	F27=0	F26=0	F25=0	Speaker ID = 0x02 (**FC**)				
16	X = 0.0 dm: **0.0** (0x00)							
17	Y = 28.0 dm: **1.0** (0x40)							
18	Z = 5 - 8.5 = -3.5 dm: **-0.21** (0xF3)							
19	F17=0	COORD=1	Active=1	Channel Index (0 to 31) = **0x05**				
20	F27=0	F26=0	F25=0	Speaker ID = 31 (0x1F)				
21	X = -30.8 dm: **-0.75** (0xD0)							
22	Y = -9.2 dm: **-0.33** (0xEB)							
23	Z = 3.5 dm: **0.2** (0x0D)							
24	F17=0	COORD=1	Active=1	Channel Index (0 to 31) = **0x06**				
25	F27=0	F26=0	F25=0	Speaker ID = 31 (0x1F)				
26	X = 35.9 dm: **0.88** (0x38)							
27	Y = -16.6 dm: **-0.6** (0xDA)							
28	Z = -1.6 dm: **-0.1** (0xFA)							

Table 207 - Example 5, Second SLDB indicating 3.1 with dual FC + Four Arbitrary Speakers

	bits							
Byte #	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	CTA Tag Code (0x07)			Length of following block payload (bytes) = 11 (0x0B)				
2	CTA Extended Tag Code (0x14)							
3	F17=0	COORD=1	Active=1	Channel Index (0 to 31) = **0x07**				
4	F27=0	F26=0	F25=0	Speaker ID = 31 (0x1F)				
5	X = -24.3 dm: **-0.6** (0xDA)							
6	Y = 9.2 dm: **0.33** (0x15)							
7	Z = 16.2 dm: **0.95** (0x30)							
8	F17=0	COORD=1	Active=1	Channel Index (0 to 31) = **0x08**				
9	F27=0	F26=0	F25=0	Speaker ID = 31 (0x1F)				
10	X = 32.7 dm: **0.8** (0x33)							
11	Y = 14.0 dm: **0.5** (0x20)							
12	Z = 15.4 dm: **0.9** (0x3A)							

The Source, if feeding all speakers indicated in SLDs with nine L-PCM channels, would send the following five channel pairs with the channels in order of the Channel Index in the SLDs, and the tenth, inactive channel silent:

FL/FR, LFE1/(FC 0.0,1.0,0.79), (FC 0.0,1.0,-0.21)/(-0.75,-0.33,0.2), (0.88,-0.6,-0.1)/(-0.6,0.33,0.95), (0.8,0.5,0.9)/NULL

The first and second channel pairs carry the three standard speaker signals and the signal for the upper Front Center, and the remaining channel pairs the signals for the lower Front Center and the four speakers at the arbitrary locations.

With that, it would carry an Audio InfoFrame as shown in Table 208:

Table 208 - Example 5 Audio InfoFrame for 3.1 with dual FC + Four Arbitrary Speakers

InfoFrame Type Code	InfoFrame Type = 0x04							
InfoFrame Version Number	Version = 0x01							
Length of Audio InfoFrame	Length of Audio InfoFrame = 10							
Data Byte 1	CT3=0	CT2=0	CT1=0	CT0=0	F13=0	CC2=0	CC1=1	CC0=0
Data Byte 2	F27=0	F26=0	F25=0	SF2=0	SF1=0	SF0=0	SS1=0	SS0=0
Data Byte 3	F37=0	F36=0	F35=0	CXT4=0	CXT3=0	CXT2=0	CXT1=0	CXT0=0
Data Byte 4	1	1	1	1	1	1	1	1
Data Byte 5	DM_INH	LSV3	LSV2	LSV1	LSV0	F52=0	LFEPBL1	LFEPBL0
Data Byte 6	**CID07=1**	**CID06=1**	**CID05=1**	**CID04=1**	**CID03=1**	**CID02=1**	**CID01=1**	**CID00=1**
Data Byte 7	CID15=0	CID14=0	CID13=0	CID12=0	CID11=0	CID10=0	CID09=0	**CID08=1**
Data Byte 8	CID23=0	CID22=0	CID21=0	CID20=0	CID19=0	CID18=0	CID17=0	CID16=0
Data Byte 9	CID31=0	CID30=0	CID29=0	CID28=0	CID27=0	CID26=0	CID25=0	CID24=0
Data Byte 10	F107=0	F106=0	F105=0	F104=0	F103=0	F102=0	F101=0	F100=0

10 ERRATA

The fifth paragraph of Section 7.4 of CTA-861-I [A] is modified, as shown below, to correct a section reference:

> The CTA Data Block Collection shall conform to the format shown in the example in Table 65. The order of the Data Blocks is not constrained. It is also possible to have more than one of a specific type of data block if necessary to include all of the descriptors needed to describe the Sink's capabilities. Section 7.6 details which data blocks may have multiple instances, and which may not.

Table 50 in Section 6.9 of CTA-861-I [A] is modified, as shown below:

Table 50 - Dynamic Range and Mastering InfoFrame

InfoFrame Type Code	InfoFrame Type = 0x07					
InfoFrame Version number	Version = 0x01					
Length of InfoFrame	Length of following Dynamic Range and Mastering InfoFrame					
Data Byte 1	F17=0	F16=0	F15=0	F14=0	F13=0	Transfer_Function (3 bits)
Data Byte 2	F27=0	F26=0	F25=0	F24=0	F23=0	Static_Metadata_ Descriptor_ID (3 bits)
Data Byte 3	Static_Metadata_Descriptor					
…	…					
Data Byte n	…					

The first paragraph of Section 7.6 of CTA-861-I [A] is modified, as shown below, to correct formatting and other minor issues:

> As noted in Section 7.4, EDIDs may contain more than one instance of a specific type of data block in its CTA Extension(s). This might be desirable for many reasons: the need to stretch the information across Extension boundaries; the desire to place "advanced" features in a second Extension that legacy devices will not be exposed to; or to make adding and subtracting Short Audio Descriptors easier in firmware. Table 123 lists the rules for allowing multiple instances of all Data Blocks and how multiple instances are handled.

Consumer Technology Association Document Improvement Proposal

If in the review or use of this document a potential change is made evident for safety, health or technical reasons, please email your reason/rationale for the recommended change to standards@CTA.tech.

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